

PROJECT AIMS AND BACKGROUND

Aims: Machine learning has made massive strides in the last decade, providing solutions to many real-world problems. Many of its applications are to text — for instance ChatGPT — or similar media. But text has aspects such as large-scale structure that can make some techniques either impractical, inaccurate or otherwise inadequate. ChatGPT, for example, generates semantically-coherent text, particularly on short passages, but fails to generate realistic long-form text [1, 2] (ChatGPT’s authors claim it considers up to 3000 words [3], which is remarkable, but it is easy to ‘trick’ it into generating contradictory or confused statements). More generally, modern recurrent deep learning architectures like LSTMs, have been *proven* to miss a property called Long-Range Dependence (LRD) [4]. This is a potential emerging crisis for machine learning models, because almost all human behavioural datasets contain LRD. This proposal aims for **new tools to analyse and synthesise LRD in non-numerical data** in order to **redress the problems created for modern machine learning methods**.

Loosely, LRD means long-range correlations in data such that the large-scale structure of the data is as important as the short-range structure. LRD is associated with the equally important ideas of statistical self-similarity and multi-scale burstiness. Together they form a set of properties that make such sequences *qualitatively* different from the Short-Range Dependent (SRD) processes that are more common in models. Nominal, categorical variables are those that take a discrete set of values with no inherent ordering: common examples being colours or a type of fruit (apples and oranges, for instance). Here we use the more colloquial terminology that such variables are Non-Numerical (NN).

Given those definitions (to be refined later) this project is motivated by four key points.

1. **LRD is common:** Strong evidence for LRD is frequently found in real time-series data, particularly when that data is associated with natural phenomena. Hurst [5] first noted the phenomena in Nile River flood data in the early 50s, but it has also been seen, for instance, in Internet traffic [6, 7], financial data [8, 9], climate [10] and many other contexts. Estimation of this phenomenon is a topic of long-term study, and there are still recent additions being made to the literature [11], but it is found particularly often in datasets resulting from human behaviour.
2. **LRD is studied primarily on numerical data:** The vast majority of work on LRD time series has concerned numerical data, for instance, sizes of floods [5], or volumes of Internet traffic [7]. However, there is a very large set of interesting time series where the data is non-numerical. There are several examples presented later, but for some initial intuition consider text made up of a series of alphabetic symbols. Most analysis of such data has been performed by converting to numerical values, but (non-linear) conversion of time-series can distort the nature of correlations, and in particular change LRD characteristics of a sequence.
3. **LRD makes processes hard to analyse:** The presence or absence of LRD leads to a qualitative difference in the manner in which data must be analysed. Some standard statistical intuitions, such as those arising from the Central Limit Theorem, do not apply to LRD time series. And yet, the vast majority of AI and Machine Learning (ML) being applied, for instance to natural language, makes assumptions that are tantamount to the data being SRD.
4. **LRD is important:** The presence of LRD can lead to qualitative differences in mechanisms needed to control processes governed by LRD input. For instance storage mechanisms (*e.g.*, queues, water-storage or inventories) show qualitatively different behaviour when fed with LRD input [12, 13]. Likewise scheduling [14], risk calculations [15] and anomaly detection [16] are affected by the presence (or otherwise) of LRD.

Those issues lead to the core aim of this project: **to develop the mathematical framework, and algorithmic tools to analyse and synthesise LRD non-numerical time series**. There is both a need for, and a lack of algorithms and understandings to apply to (Non-Numerical) NN-LRD datasets such as natural language. Initial work has commenced, for instance, in the study of gene expression [17] or natural language [18], but currently there has been no systematic effort focused on the topic.

Background: Historically the focus of time-series analysis has been numerical time series, such as prices, air temperature or an audio signal. However, there are an increasing number of large to very-large, non-numerical (NN) datasets:

1. Event logs are an increasingly important form of cyber-security and reliability data. Events, for instance in a large data centre, could contain instances of device or software failures, cyber-attacks, and associated environmental variables such as the state of the internal network.
2. The Common Vulnerability and Exposures (CVE) database is a dictionary of publicly known security vulnerabilities [19] launched in September 1999. It is the world-wide register of cyber-security vulnerabilities and exploits. A pure count of the number of such entries over time shows evidence of LRD [20], but the structured sequence of disclosures includes information such as type of vulnerability and is much richer.
3. Human genomic data has been shown to have evidence for LRD [21], though the evidence is not conclusive in part due to a lack of suitable tools for analysis.

4. Large corpora of text, *e.g.*, Project Gutenberg contains 60,000 books [22]; and TVeeder [23] streams captions from TV programs shown on free-to-air TV in Australia. Work over many years has assumed and/or attempted to measure the existence of LRD in natural language [18, 24–28]¹.
5. Social media contains massive volumes of text, but also, contains a network of relationships that evolve over time, Networks are by their nature discrete, non-numerical data structures.

Note, the datasets we highlight are those with which we already have some experience and where there is already some evidence of LRD. We will, however, extend our analysis to other datasets as and when they are available.

The analysis of NN-LRD data is both important and lacking. And it is important that we correct this. The ability to analyse these datasets and in some cases synthesize them is becoming increasingly important. There are too many applications to list here, but for a subset see:

- Anomaly and change detection, *e.g.*, to detect problems.
- Natural Language Processing (NLP) algorithms, *e.g.*, sentiment analysis.
- Data synthesis, *e.g.*, cyber-deception for creation of more realistic honeypots.
- Fake news and botnet detection.

Apart from the many application, the deeper scientific component of this project concerns the desire to validate the findings of LRD in NN datasets by providing more accurate and robust estimators, and to discover if LRD exists in new datasets.

INVESTIGATORS/CAPABILITY

CI Roughan's research has encompassed mathematical modelling, measurement, and management of networks, primarily computer networks such as the Internet. He has worked with industrial giants such as AT&T (USA), and Ericsson (Sweden). His work in this area, for instance, helped change the way AT&T runs its network, saving them tens of millions of dollars each year. And it has resulted in election to Fellow in both the ACM and IEEE, and many invitations to present many keynotes. Prof. Roughan will be lead CI on the project, contributing 0.2 FTE. He will provide research leadership for the project, and expertise primarily related to a long-term theme in Prof Roughan's research, which has concerned analysing the statistical properties of Internet traffic, and one important strand of that research has concerned LRD. Several large projects funding end in 2023 or soon afterwards, so Prof Roughan will have capacity to take on a major new project. As current director of the TRC centre in the university, Prof Roughan will also provide connections to industry to transition research results into impactful applications.

PI Willinger is one of the founders of internet measurement as a research domain, and his seminal works on LRD in network traffic have collective citations measured in 10's of thousands. He currently works as Chief Science Officer for Niksun, a company that works on cyber-security tools, in particular employing ML approaches, and so PI Willinger provides both research strength to the project and a pathway towards real applications of ML in cybersecurity.

The CI and PI already collaborate extensively, for instance as CI/PI in ACEMS (the recently concluded ARC Centre of Excellence for Frontiers in Mathematics and Statistics). And they collectively have a long and growing list of joint publications.

This proposal asks for funding for one Research Associate who will collect and analyse data, extend models, and develop the computational tools. They will require skills in mathematical modelling and analysis to develop mathematical frameworks; programming and algorithm skills to develop new analysis algorithms; and data collection and statistical analysis abilities. Given the range of skills and experience required from this person and the range of tasks to be completed, this will be a full-time project for three years.

The project also asks for HDR student stipends to support graduate students to work on components of this project. There are several components of this project that are ideal for a graduate project over a range of topics, and for students from any of mathematics, physics, statistics or computer science backgrounds, specifically: analysis of one of the data sets or development of a specific synthesis or estimation algorithms.

We will also seek to attract independently funded students to the the project, with the aim of attracting two additional postgraduate students per year. This is a conservative estimate based on the number of new students per year attracted by CI Roughan – he currently supervises 10 graduate students so this goal is realistic.

PROJECT QUALITY AND INNOVATION

Tools such as GPT-3 [1, 2] have made remarkable strides in generating human-like text. Such tools are very good at generating or analysing local structures such as valid sentence syntax and stylistic features of a text. But these tools struggle to create or appreciate the larger-scale structure that real human texts contain. For instance, analysing plot structure has an extremely long history, arguably dating back to Aristotle (the Poetics). In modern (Hollywood) filmmaking the notion of the “three act structure” characterised by rising tension followed by resolution is codified/popularised by

¹Note the term long-range dependence (and related terms) are often used colloquially rather than mathematically in the context of natural language. The two meanings are linked, however, and the former is indicative of the phenomena that we wish to quantify in this project.

Field [29]. Taxonomies of stories more generally have been debated throughout the 20th century, with characteristic numbers of plots/stories found by authors apparently decreasing over time, from thirty-six in 1916 [30], to twenty in 1993 [31], to seven by 2004 [32], and six in 2016 via a data-driven approach [33]. These large-scale structures are missing from most attempts at automatically generating or understanding texts, and in fact Belletti *et al.* [4] have proved that common neural architectures cannot incorporate LRD.

Long-range structure and statistical self-similarity are features of almost all human activities as well as many other natural phenomena. In the modern context, a boom in research in the topic occurred after it was noted by Leland *et al.* [34] in Internet traffic data. Recent work, unsurprisingly, identifies long-range structure in human-generated text as well as other media such as music [18, 27, 28], though we argue that their measurement methodology can be improved.

Long-range structure has profound importance for statistical estimation. When long-range correlations are present in a time series standard statistical intuition fails: for instance, the Central Limit Theorem does not apply. Therefore long-range dependence can impact every aspect of text measurement and generation. Most recently, researchers have started to realise that such structure should influence the design of neural architectures applied to such data [4, 18, 35].

Neural architectures for studying language and other media is a booming research field that we could not hope to describe within the space limitations of this proposal. The important facts to note, however, is that although many such architectures attempt to build some longer structures, these are typically limited to very finite distances both by the need to force stability into an innately non-linear recurrence relationship [4].

Long-range dependence in time series is strongly linked to the twin ideas of burstiness and statistical self-similarity. While these have been extensively studied in numerical time series, there is less understanding of the phenomena in text, which is by nature non-numerical. Models that incorporate self-similar, hierarchical structures should be able to mimic structure to arbitrary extents, but there are many unknowns. For instance, open questions include: how to build computationally-practical, stable neural architectures that support such structure; and how to combine textual content with temporal process models (*e.g.*, Hawkes processes, which Zhu *et al.* [35] have attempted to combine with deep learning methods such as transformer network). New approaches such as EvolutiveRNNs and Compressive Transformers have been proposed as solutions, but there is still a large gap in understanding between the theoretical models and properties and practical generation models. This project aims to bridge that gap.

The fundamental data to be considered in this project are non-numerical (nominal categorical) discrete-time series. The typical representation would be a sequence of symbols, *e.g.*,

abcacc...

Long-Range Dependence has been defined through a number of measures. We focus on the following set of definitions where the autocovariance function (ACF) of a zero-mean stationary sequence X_n is $\gamma_k = \text{Cov}(X_i, X_i + k)$, and the spectral density is the Fourier transform of the ACF. There are two common (and equivalent) definitions of LRD.

Definition 1 (LRD: time domain). *A sequence is said to be LRD if and only if sums of the tail ACF diverge, i.e., $\sum_{k=m}^n |\gamma_k| \rightarrow \infty$ as $n \rightarrow \infty$, but usually this is made more specific by specifying that the tail of the ACF takes power-law form, i.e., as $k \rightarrow \infty$ we require that $\gamma_k \sim c_\gamma |k|^{-\beta}$ for $\beta \in (0, 1)$, [36].*

Definition in terms of summation is useful because it expresses that no matter how far into the tail we progress, the correlations never diminish to the point where they are inconsequential, but the power-law decay is more usually used because it leads to a parameter, β , that measures the ‘strength’ of LRD.

Definition 2 (LRD: frequency domain). *LRD is associated [37, 38] with a pole in the frequency domain at frequency 0. Typically this is called $1/f$ noise, though in many cases the spectral density has the asymptotic form of $1/f^\alpha$ with $\alpha \in (0, 1)$ as $f \rightarrow 0$.*

There are several other similar definitions, for instance in the wavelet domain. However, in the context of some common stochastic processes (*e.g.*, point-processes) the previous definitions are not directly applicable, and hence an alternative is typically used, *e.g.*, [39] based on counting functions. A typical version involves the function $N_S(t)$ that counts the number of visits to a set of states S up to time t , *e.g.*,

Definition 3 (LRD: counting). *A countable-state Markov chain is said to be LRD if and only if the variance of the count function grows faster than linearly, i.e., $\limsup_{n \rightarrow \infty} \text{Var}[N_S(n)]/n = \infty$.*

This definition is analogous to divergence of sums of the autocovariance function and is related to an equivalent parameterisation. It also leads to a definition in terms of recurrence times [39]:

Definition 4 (LRD: recurrence times). *A stationary, positive-recurrent, irreducible, aperiodic Markov chain is long-range dependent if and only if the return time random variable $T_{ii} = \inf\{n \geq 1 \mid X_n = i \text{ given } X_0 = i\}$ has infinite second moment for all states i .*

This last definition is the first that also applies to (finite-state) NN data, so it is a useful branching point, but it has limitations, *e.g.*, for highly skewed (Zipf-like) distributions, and there are many other properties of interest, for instance:

Definition 5 (Statistical self-similarity). *A process is said to be statistically self-similar [34] if aggregated and appropriately scaled versions of the process have the same statistical properties. Formally, define the aggregated process of a discrete-time process as $X_k^{(m)} = \sum (X_{km+1} + X_{km+2} + \dots + X_{km+m})/m$ then a process is self-similar if and only if $\{m^{-H} X_k^{(m)}\} \stackrel{d}{=} \{X_k\}$ for all m , where $H \in (1/2, 1)$ is called the Hurst parameter and $\stackrel{d}{=}$ refers to equality of all finite-dimensional distributions.*

Typically we are interested in whether self-similarity holds asymptotically for large m and for second-order statistical distributions, which brings us back to the limiting behaviour of the ACF. The alternative definitions are useful in this context because they each provide means towards synthesis and analysis of LRD processes [34]. They are linked by parameters $\beta = 1 - \alpha$ and $H = (\alpha + 1)/2$, which are therefore crucial characterisations of the phenomena of LRD and self-similarity, and the primary parameters that we wish to estimate.

There are other definitions of LRD (such as the rescaled range analysis of Hurst himself) that have led to estimators in the more conventional case of quantitative data [36] but not all of these are equivalent. Our goal in describing the alternatives here is to illustrate that there are many approaches to extending that analysis to NN data. There are also many extensions to continuous-time processes, but these are less applicable to the type of data we consider here, *e.g.*, text. Note also that we can meaningfully characterise processes with parameters outside the ranges given, but they are no longer stationary LRD processes.

There are however many challenges in adapting LRD techniques to the domain of interest.

1. Many NN datasets have a large number of symbols, *e.g.*, consider NLP, where words are considered as the symbols. The number of symbols in a text is then the vocabulary size which can be measured in tens of thousands in some cases. The number of graphs of a given size n is $2^{n(n-1)/2}$, and this can be an even larger number than that of vocabulary. The size of the state space can lead to computational and numerical problems, but also touches on the *curse of dimensionality*.
2. The distribution of the occurrences of signals can be wildly variable. Again, if words are the symbols then their frequency is commonly approximated by the Zipf or Zipf-Mandelbrot distribution, which is heavy-tailed. Skewed distributions mean that some symbols will be only rarely seen, and hence estimation must focus on common events while including rare events. For instance, analysis of recurrence times could be hazardous if blindly applied.
3. LRD is often associated with other forms of statistical complexity, *e.g.*, multifractality, etc. and these may complicate the problem.
4. LRD analysis typically requires very long sequences, which can be available in the case of, for instance, Internet traffic data, where it is a matter of how long data is observed, but may be more difficult in the context of a fixed length text.
5. Non-stationarity in the underlying sequences can be mistaken for LRD and visa-versa [40].

Within the context of these challenges, we present a set of approaches and aim to test and compare these.

Project details

Task 1: Develop and Extend Methods for Generating LRD sequences of symbols:

A key aim of this project is to analyse real data. However, we lack *ground-truth* for real data and so evaluations should also use synthesised data where we have control over parameters, can generate data of arbitrary size and where we know the true parameters of the data.

The synthesis of quantitative LRD time series is well documented. The approaches might be broadly classified as:

1. **Property based:** that are based on creating the property of LRD or self-similarity, where the model is not postulated to have any particular relation to the “physics” of the system. For instance:
 - Time domain methods based on self-similar models (Definition 5), *e.g.*, fractional Brownian motion (fBM) generation through recursive Random Midpoint Displacement (RMD) [41].
 - Frequency-domain methods aimed at creating the spectral properties of LRD models (Definition 2) [42, 43].
 - Combined time-frequency domain techniques, for instance, using wavelets, *e.g.*, [44].
2. **Model based:** where the model (usually a stochastic process model such as a Fractal Renewal Process, Heavy-tailed On/Off processes, Fractal Shot Noise Processes, or superpositions of these [7, 45, 46]) is postulated as an abstract model of the underlying system.

Note the above references are just a sample of the many efforts to generate LRD processes. This was a rich field of research in the 90s to early 2000s. Each of these approaches has advantages or disadvantages for particular applications. Speed is often required because of the length of sequences needed, but speed usually comes at the cost of approximation or inflexibility. Memory requirements are also potentially costly for very long sequences. There are also subtle flaws in some approaches, for instance, in [47] we found that FRPs could show a range of scales much more limited than the length of the sequence being generated, therefore limiting their ability to demonstrate true LRD.

The first task of this project is to investigate and test various approaches for generation LRD symbol sequences. As above we broadly classify these into property- and model-based approaches, and there is a rich variety of both.

1. **Property based:** underlying the time-series (and frequency domain) techniques is the idea of ordinal (often Gaussian) data. However, we can easily adapt this to generate non-numerical data by quantisation. Most simply when a numerical sequence falls inside a given set of levels it generates a particular symbol. More generally we could adapt the latent Gaussian approach to derive a categorical sequence from Gaussian processes by extending [48].

This brand of approaches is appealing because we can reuse the existing algorithms for generation of the underlying sequences. However, they have one potential flaw which is that it is difficult to generate a sequence with a desired short-range structure. For instance, in the context of NLP, symbol bigrams (pairs of symbols) are structured – for instance in English the letter ‘q’ is usually followed by a ‘u’. Alternative approaches to allow for such a structure might include using the underlying fractal process to drive a Markovian state machine, where probabilities of the next states are controlled through the generation process, and the state machine enforces local structure, or hybrid time/frequency methods based on wavelets.

2. **Model based:** There are several models that can be used to create LRD symbol sequences:

- Infinite-state-space Markov models can exhibit LRD. The simplest example is, perhaps, the unbiased random walk on the integers. Recurrence times in such processes can be heavy-tailed, but it can be hard to control the parameters of such a process.
- The Linear-Additive Markov Process (LAMP) [49] can be adapted to generate sequences with arbitrary auto-correlation with finite states. The Custom Decay Language Model (CDLM) of Singh *et al.* [26] is a variant.
- Doubly-stochastic models have been used to generate, for instance, point processes exhibiting LRD such as Fractal-Shot-Noise processes (FSNP) [50], but might equally be adapted to NN processes, for instance, by using an inhomogeneous Markov chain with probabilities driven by a heavy-tailed On/Off process.
- A Fractal-Symbol-Sequence (FSS) is a novel proposal (of this project) whereby the time at which each intended symbol appears is controlled by an independent fractal point process such as a FRP or FSNP.

Model based approaches are amenable to combination and extensions through superposition, filtering, and various other transformations. They also have the advantage of being linked to the root causes of LRD.

We will also explore purely data-driven approaches such as Variational autoencoders (VAE).

Generation techniques will be tested through three approaches:

1. We will measure the techniques by linking to Task 2 (estimation) as quantified by, for instance, the range of scales over which LRD can be observed in the model.
2. We will quantify a method’s control of short-range characteristics such as its marginal statistics.
3. We will derive the computation and memory complexity of the techniques.

Task 2: Develop and Test Estimators:

Many approaches have been used for the estimation of LRD parameters in the context of quantitative data, *e.g.*, see [51]. In this task, we seek to develop a set of similar estimators for non-numerical data. This is a harder task than generation.

Here we propose a starting list of estimators. The project will implement, test and extend where needed.

1. Conversion to a tradition numerical sequence: *e.g.*,
 - Map the i th symbol to the value v_i and then apply existing estimators. This has the advantage of reusing existing insights, but the disadvantage that it ignores the fundamental nature of the data. Examples of this include [21], which maps DNA sequences to values, and applies a wavelet-based estimator to the result.
 - Map into a *count-sequence*, *e.g.*, a vector-valued, discrete-time sequence, where each terms $v_i(t_j)$ counts the number of occurrences of symbol j during time interval (t_j, t_{j+1}) . Then apply existing estimation techniques. This has the advantage of reusing existing insights and incorporating the nature of the data, but requires the LRD estimator to apply to vector-value sequences (most existing methods are designed around scalar-valued sequences).
 - Map using a trained neural network (*e.g.*, a VAE). This might work the best out of all conversions in terms of retaining short-term structure, but non-linear transformations can alter LRD properties [52].

2. Direct approaches based on time-series definitions: *e.g.*,
 - Calculate the auto-Cramer-V statistic and use regression on a log-log plot to estimate the decay of the autocorrelation. Singh *et al.* [26] use a related statistic, the co-occurrence rate.
 - Calculate a similar but less computationally complex statistic, such as entropy rate that depends on the underlying correlation structure, and use the convergence properties of this statistic.
3. Direct approaches based on model-based definitions: *e.g.*,
 - Exact estimators: in a small number of cases (*e.g.*, the Whittle estimator) an exact MLE has been developed to measure the parameters of a particular model. It is feasible that exact MLEs are possible for some of the generative models proposed.
 - Approximate estimators: Bayesian computational techniques have recently been applied to ARFIMA models [11]. It is likely that similar approaches can be adapted to some non-numerical models.
 - One definition of LRD in non-numerical data concerns recurrence times (see Definition 4). Measuring recurrence times in the process, and hence the parameters of their tail using heavy-tailed estimators (*e.g.*, the Hill estimator) provides another approach.

It is not obvious which approaches will work best. In fact, it is likely that no single approach will suit all circumstances. MLE estimators are ideal in the case where the model fits the data well. However, MLEs, while they have tremendous advantages also suffer from the problem of model-errors [53], and in particular it should be noted that LRD in real data is usually an asymptotic property, hence exact LRD models rarely exactly match data.

Also, there are particular challenges in different types of data: the number of symbols if we consider an alphabetic representation of natural language is of the order of 60-70 symbols (lower and upper case letters and punctuation marks) and the distribution over these, though not uniform, is not extremely skewed, but when we consider the data as a series of words, then the number of symbols is the vocabulary size (which might typically be measured in thousands or tens of thousands) and the distribution is so skewed it is usually represented as a Zipf or Zipf-Mandelbrot distribution.

Implementation of some estimators will be non-trivial. Some might require pattern matching over very long sequences (millions of symbols); others are based on calculating statistical measures where naive approaches could blow up either in memory or computation.

Testing also needs to be conducted carefully. Ideal estimators include diagnostics (residual estimates, or confidence bounds), but not all of the above come with such by default. Methods such as bootstrap can be used to create, for instance, confidence intervals, but that could multiply the amount of computation substantially.

Methods of testing of estimators include:

1. Errors (bias and variance) as a function of the amount of data.
2. Diagnostic ability.
3. Robustness to (i) modelling errors (*e.g.*, the fact that LRD is usually asymptotic), or (ii) non-stationarity in the data, which can fool many estimators into see LRD in SRD data [40].
4. Computational and memory complexity.

Task 3: Analysis of Real Data

We intend to apply the best approaches of Task 2 to real data, starting with the following:

Dataset 1 – Online Social Network (OSN) data: OSN analysis is an existing strength at the University of Adelaide. We will start by using an existing Twitter decahose (10% of all public tweets) dataset that we have access to, which was used in the *hedonometer.org* project. CI Roughan has access to this dataset comprising 15 years of public Twitter data. We will then extend and link this data to other OSNs, for instance to Reddit. The online social network platform Reddit has an inherently hierarchical networked structure. Our preliminary analysis suggests that the Matthew effect exists on Reddit, *i.e.*, the early mover, or rich get richer effect (which is linked to LRD).

Dataset 2 – Cyber: Problems applying ML to cybersecurity problems are well documented, *e.g.*, see [54]. The Common Vulnerability and Exposures (CVE) database is a dictionary of publicly known security vulnerabilities [19] launched in September 1999. It is the worldwide register of cyber-security vulnerabilities and exploits. A pure count of the number of such entries over time shows evidence of LRD [20], but the structured sequence of disclosures included information such as type of vulnerability and is much richer. We will seek to find hidden structure in this data, but more importantly, use it to ensure that other measures of this data (for instance studies of the correlation between types of disclosures and real-world exploitation) use correct statistical procedures.

Dataset 3 – Media and media networks: The world’s modern media are one of the most studied forms of human-originated data. We plan to study both natural language texts as well as other forms of media. In addition, media networks have been analysed from early in the current era of network science, an early example being the collaboration graph between actors. Symbol sequences are one example of non-numerical data, but they are comparatively well understood. A more complicated dataset is an evolving network. In that context, even the idea of “correlation” is not

as well defined, or at least is susceptible to multiple definitions depending on the type of network (directed/undirected, weighted/unweighted, labelled/unlabelled) and construction, *e.g.*, De Bruijn.

Dataset 4 – Packet-loss data: Packet-loss measurements in the Internet are used to assess network performance. There is ample evidence of LRD in Internet data (for instance see [34] and its nearly 10,000 citations) but the correlations in packet losses are rarely accommodated, and even our own work only allows for relatively short-tailed correlations [55]. We plan to reprise our existing analysis taking into account LRD in the loss process.

We do not plan to limit our work only to these datasets, but these are data with which we already have immediate access to large corpora and extensive experience in working with the data.

Task 4: Develop Applications of the Techniques and Models

The ultimate goal of this work goes beyond satisfying our scientific curiosity. We would like to be more able to work on NN-LRD data. Potential applications include

1. The statistical properties of LRD sequences break many standard statistical intuitions, for instance, the Central Limit Theorem does not apply to such sequences, and hence statistical analysis of such data requires additional care. For one simple example, consider that confidence intervals for estimated properties of such sequences will typically be much wider than for SRD processes, but in some cases, estimates (*e.g.*, predictions) can be tighter [16]. Developing new statistical techniques to apply to such sequences will be a major outcome of this project. A simple example is forming accurate confidence intervals for loss-rate measurements in packet networks [55] as noted above.
2. Data synthesis has applications other than generating test data: (i) creating artificial datasets to protect against privacy violations a topic that has received recent interest in health research [56], or (ii) creating artificial "cyber-deception" datasets for use in honeypots to detect and provide probative evidence against hackers [57]. As far as we are aware neither of these applications has used LRD-aware techniques, and yet both seek to emulate human-driven processes.
3. Detection of anomalies, fake data, malicious actors and other problematic sources within complex data sequences such as Twitter or news media. The current war in Ukraine illustrates how fake news is used to dis-inform and confuse the public about a horrific situation. Detection of problematic sources would decrease the 'noise' and make a rational discourse more feasible. Fake sources often mimic human statistical behaviour to hide, but it is harder to fake LRD simply because it is less well understood and thus it provides an angle from which to find fakers because anomalies stand out more clearly [16].
4. Sequences such as the disclosure of CVEs lead to workload for organisations that must guard against the disclosed vulnerabilities. Workload processes arising from LRD processes have different characteristics and if dealt with carelessly could result in inefficient or dangerously slow processing, but a little understanding can lead to simple heuristic solutions, *e.g.*, [20], that make processing such workloads more effective.
5. Machine learning seeks to perform many tasks in this space: a long-term goal in the natural language space is message understanding, but even seemingly simple tasks such as part-of-speech or entity identification are non-trivial. For instance, a common assumption in building word embeddings (often used in classification tasks) is to assume words close together are related, but if text sequences possess LRD then this assumption is weakened. On the other hand, LRD leads to potential new ways to create such embeddings. Improving the standard of machine learning techniques for the analysis of text is an important application.

One appealing approach to the last applications lies in the insights that wavelets brought to the analysis of LRD [58,59]. Space prohibits a complete review of wavelet methods in the context of LRD sequences, but there are several reasons they seem applicable:

1. The wavelet structure naturally contains self-similarity, and a choice of appropriate wavelet bases can match analysis or synthesis to the properties of the sequence being observed.
2. Wavelets can be implemented very efficiently (more efficiently than the fast Fourier transform) as a pyramidal filter bank consisting of hierarchy convolutions that should lead naturally to a generalisation to convolutional neural networks by, for instance, using a network structure where convolutions are encoded through successive banks, thus providing a bridge between recurrent and convolutional neural networks.

BENEFIT

This project has potential benefits for Australia in many respects. Apart from the scientific advances made, and the building of expertise in data science and mathematics, two crucial areas for the modern world, it will have some more tangible benefits. The applications described in Task 4 above lay out many of the intended areas in which the results will be applied, but in brief:

1. One of the factors the Australian government has identified as crucial for Australia's ongoing safety is a strong information-warfare capability [60]. The information domain is referred to as the 5th warfighting domain and it is expected that it will precede and envelop conventional warfare. Detecting malicious players in the information-scape

is a core part of the required capability. Several parts of this project contribute to that goal for instance by detecting cyber-influence and by enabling the creation of better honeypots to detect and prosecute hackers.

2. Synthetic data has many other uses. For instance, it can allow the creation of genuinely anonymised datasets for medical research and it is a core of rigorous simulations of larger systems.
3. Understanding and modelling of LRD processes can enable better workflow processing, the example presented here is in processing the large number of CVE records presented each year, which presents an increasing workload for cybersecurity professionals to deal with.

The benefits are not abstract. The University of Adelaide and the School of Mathematical Sciences have a strong history of taking mathematical research to market. In the last Excellence in Research Australia (ERA) assessment, the School received the highest possible impact rating for Applied Mathematics based on “the development of software for high-speed, reliable mobile communications networks, in partnership with industry.” The mathematical software in question was developed at the Teletraffic Research Centre (TRC) supported by Telstra, but is just one part of the work the TRC has performed for 35 years in translating mathematical research into real, bankable impact. CI Roughan is the current director of the TRC and will provide a direct connection to the industry and government partners that would find the work described useful. In particular, we have working relationships with organisations like Telstra, SA Power Networks, Fivecast, SA Health and many others, and industry projects that include topics such as influence detection and medical data synthesis.

FEASIBILITY

Many of the issues of feasibility have been addressed in the previous sections, however, to recap there are several components contributing to the feasibility of this project.

1. The CI and PI already work well together. They have worked on quite a few past grants together including several ARC Discovery Grants and one ARC Centre of Excellence. They have worked on several previous grants together, and have a long list of joint publications.
2. The critical datasets in question have already either been collected or are extensions of existing data using known measurement techniques.
3. The team members brings the right skills to the project, making it possible to attack such a widely spread series of topics the development of software for high-speed, reliable mobile communications networks, in partnership with industry. CI Roughan and PI Willinger both have experience in the mathematical analysis and synthesis of LRD processes, working with large data sets and applying mathematical approaches for industry.
4. The environment at the University for this work is ideal. There is a high-quality group in applied mathematics (rated above world standard in successive Excellence in Research for Australia results) with an energetic group of 17 graduate students working on related topics. This exists within the larger context of multiple collaborations within the University, for instance with the Australian Institute of Machine Learning and the Teletraffic Research Centre (TRC). The University is also strongly allied with the Defence Science and Technology group, with tens of millions of joint projects in play. The TRC in particular is a centre devoted to transferring mathematics to industry.

COMMUNICATION OF RESULTS

The investigators believe that research results should be communicated by publishing them in the best international journals. Publication of results in refereed journals in this area often involve long lead times, and so the investigators also believe that it is necessary to report results to the research community over a shorter time-scale by presenting papers at major international conferences and through pre-prints.

The project would also aim to provide training, in the form of support for graduate students. Apart from allowing additional progress on the project, this would enable training students in data science, and area of growing importance.

However, more than that, we see education as an important part of building capacity in industry. It is important not just to provide new tools, but also to provide deeper, longer-term insights and communicate these to industry. To that end, we will extend current courses in areas such as network science and management, which have been given to industry, and also deliver these through venues such as industry and operators groups.

In addition, we believe that publicly funded research should be publicly visible, and so have often been involved with press, radio, and TV reports based on their work (for instance, CI Roughan was interviewed for a “People of the ACM” news article); CI Roughan recorded several episodes of the “Random Sample” podcast series; and have often participated in outreach activities to high schools and the wider community. This has an additional benefit that it reinforces Australia’s understanding of the importance and relevance of mathematics to our country and the world in general.

Finally, it is intended that much of the software and tools from the project would be open sourced, to provide a resource for other researchers.

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